“Everything under heaven
is in chaos;
the situation is excellent.”



GABMs are a mess

- How do agents connect?
- How are they prompted?
- How are they orchestrated?
- How are they activated?
- What can each agent see at each turn?
- Are all users LLMs—or hybrid?
- Base or post-trained?
- Generate content—or only react?
- Create connections—or fixed network?
- Does the prompt / state update?
- Do in-simulation actions update memory / state?
- Real people—or synthetic personas?
- Is only one LLM used?
- Are notifications modelled?
- What words do we even use for these choices?

Every choice can bias—or determine—the output.

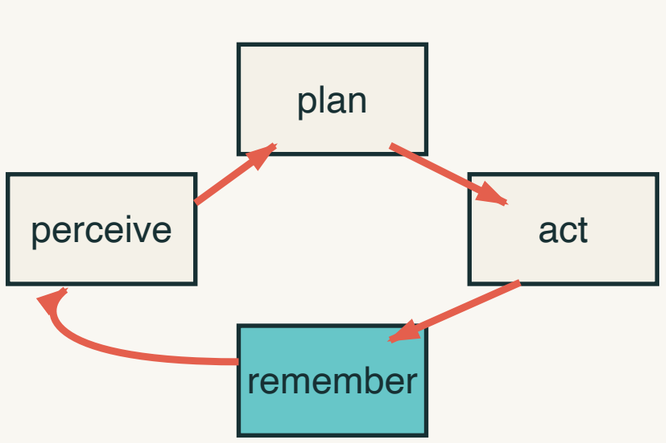
Results and claims of validity are rarely comparable.

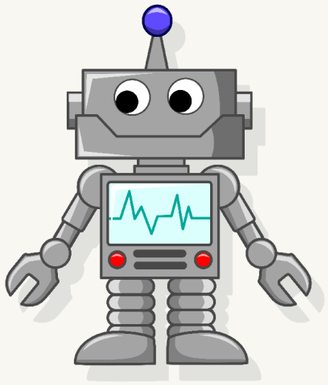
Too many degrees of freedom.

Agentic / non-agentic

LOOP

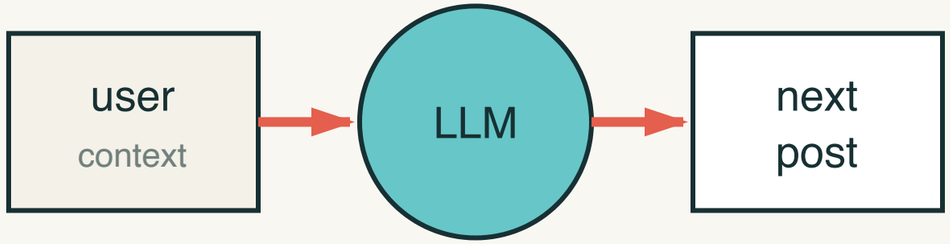
Agentic





ONE MAPPING

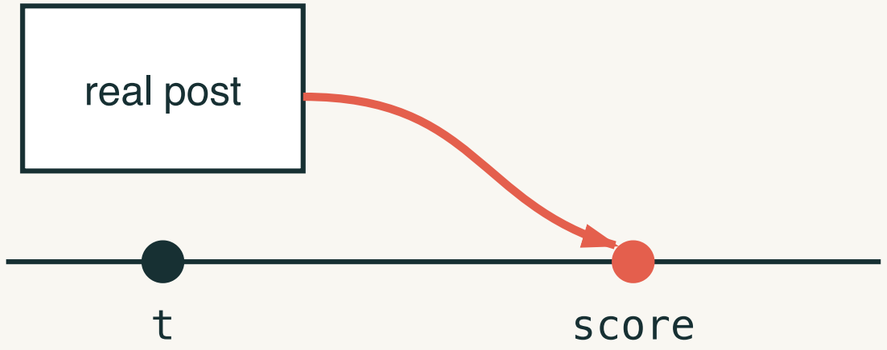
Non-agentic



Single-turn / multi-turn

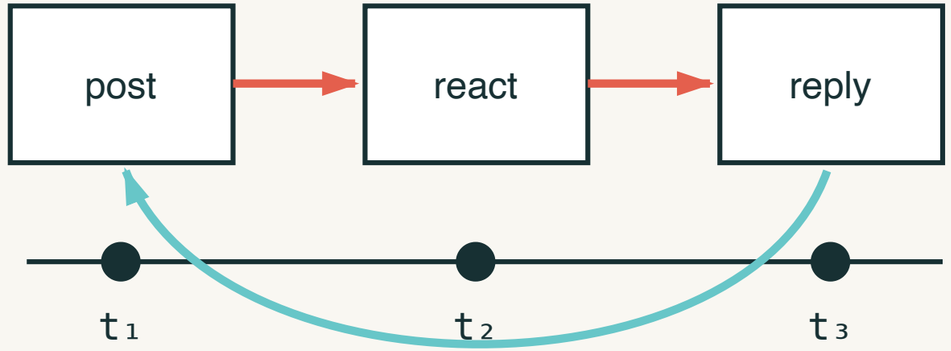
ONE OBSERVED ACTION

Single-turn



A TRAJECTORY

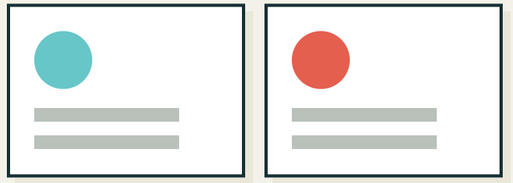
Multi-turn



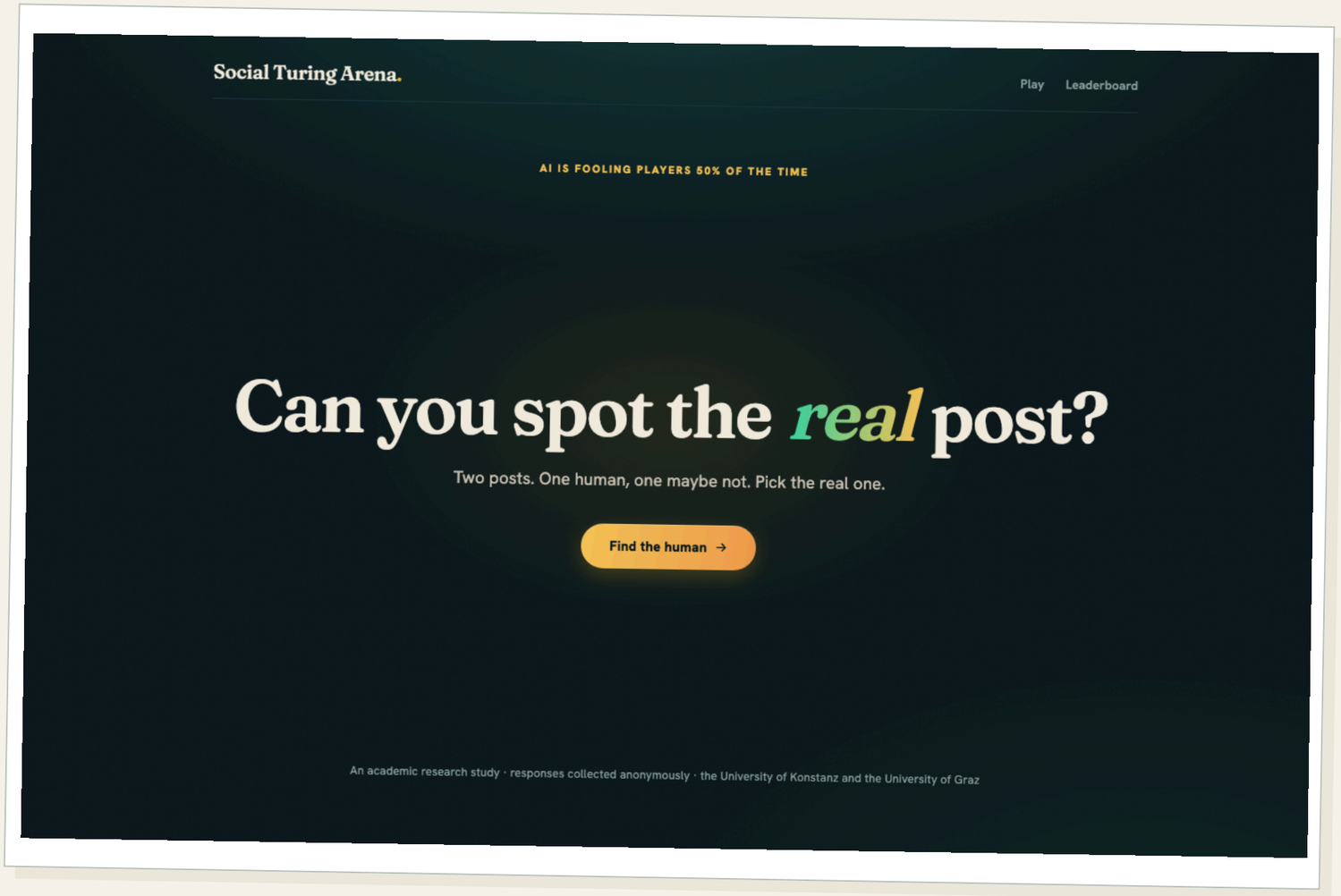
One action, or a world that feeds back?

Plausibility

real / not real

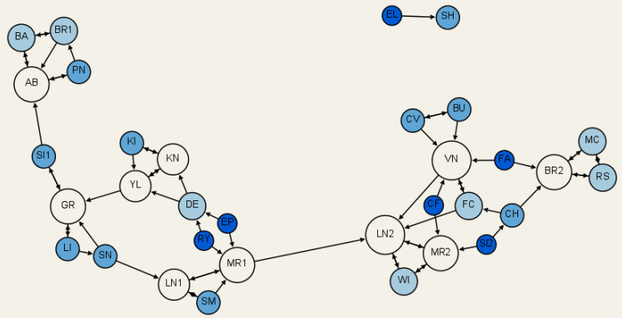


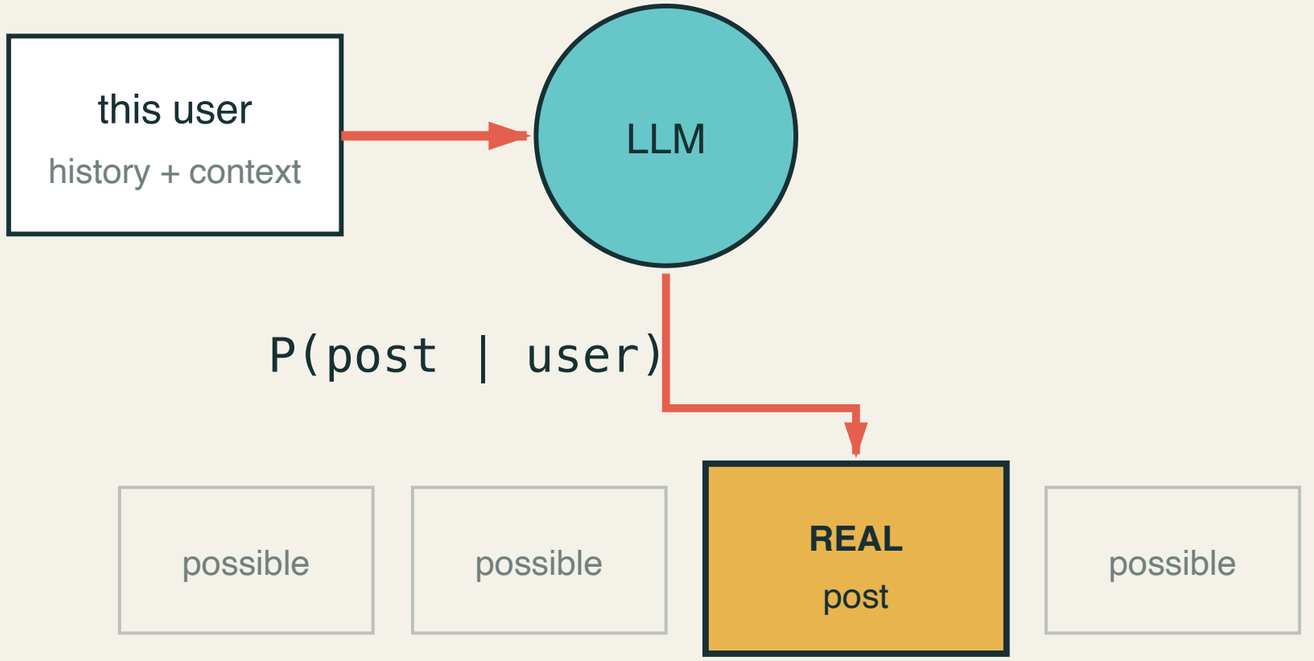
SocialSimBench



Realisticness

Persona Sweep





score what actually happened

Sociogram: Martin Grandjean, CC BY-SA 4.0 · 06 / 16

My corner

	NON-AGENTIC	AGENTIC
SINGLE-TURN	**PERSONA SWEEP**	Park et al. survey agents
MULTI-TURN	?	OASIS · SimUSER Concordia · Y Social

Perplexity

$$-\log_2 P(\text{real post}) / \text{bytes}$$

PPL → 1

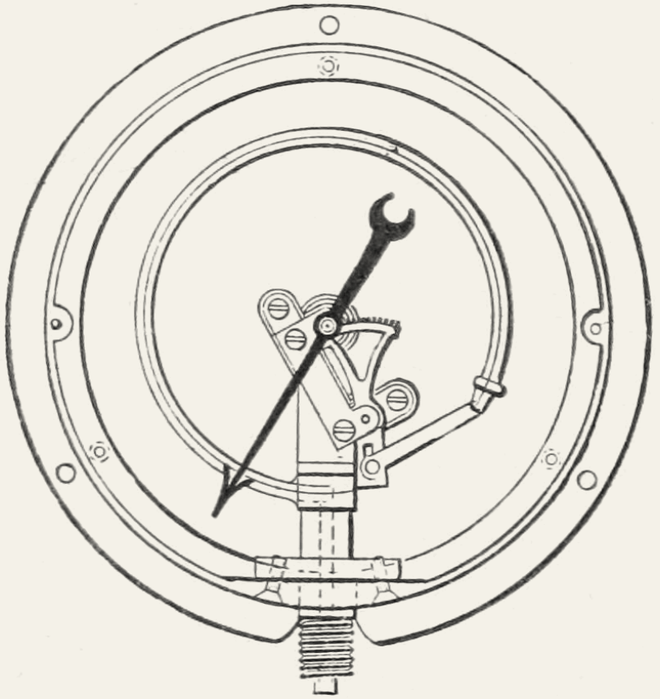
0 b/B

the real post

PPL → ∞

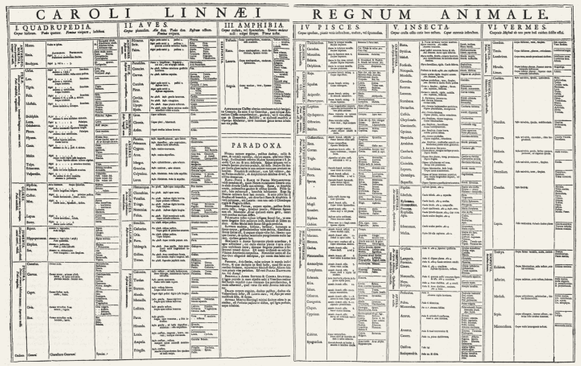
∞ b/B

no chance

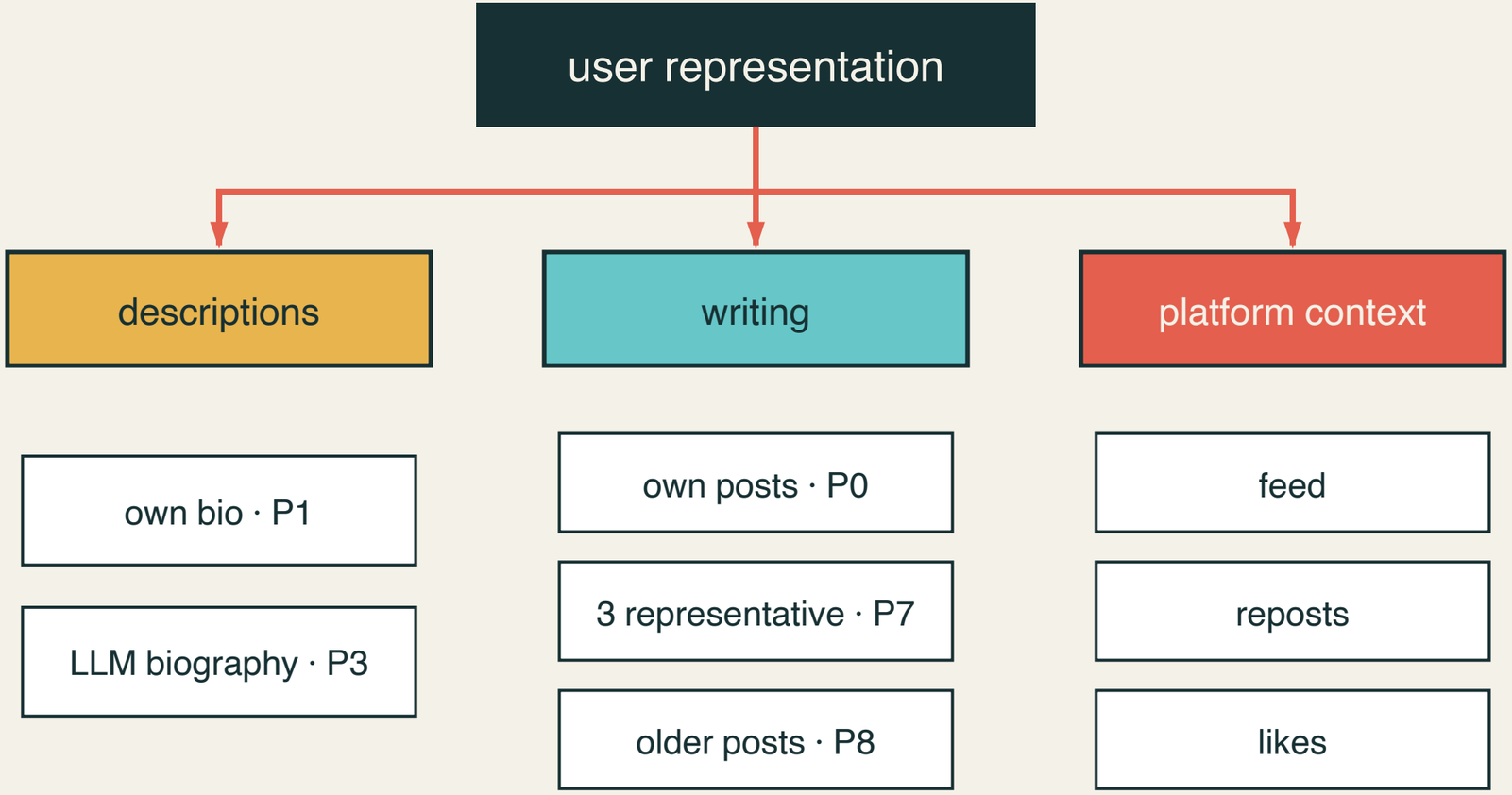


lower = better · comparable across tokenizers

Personas

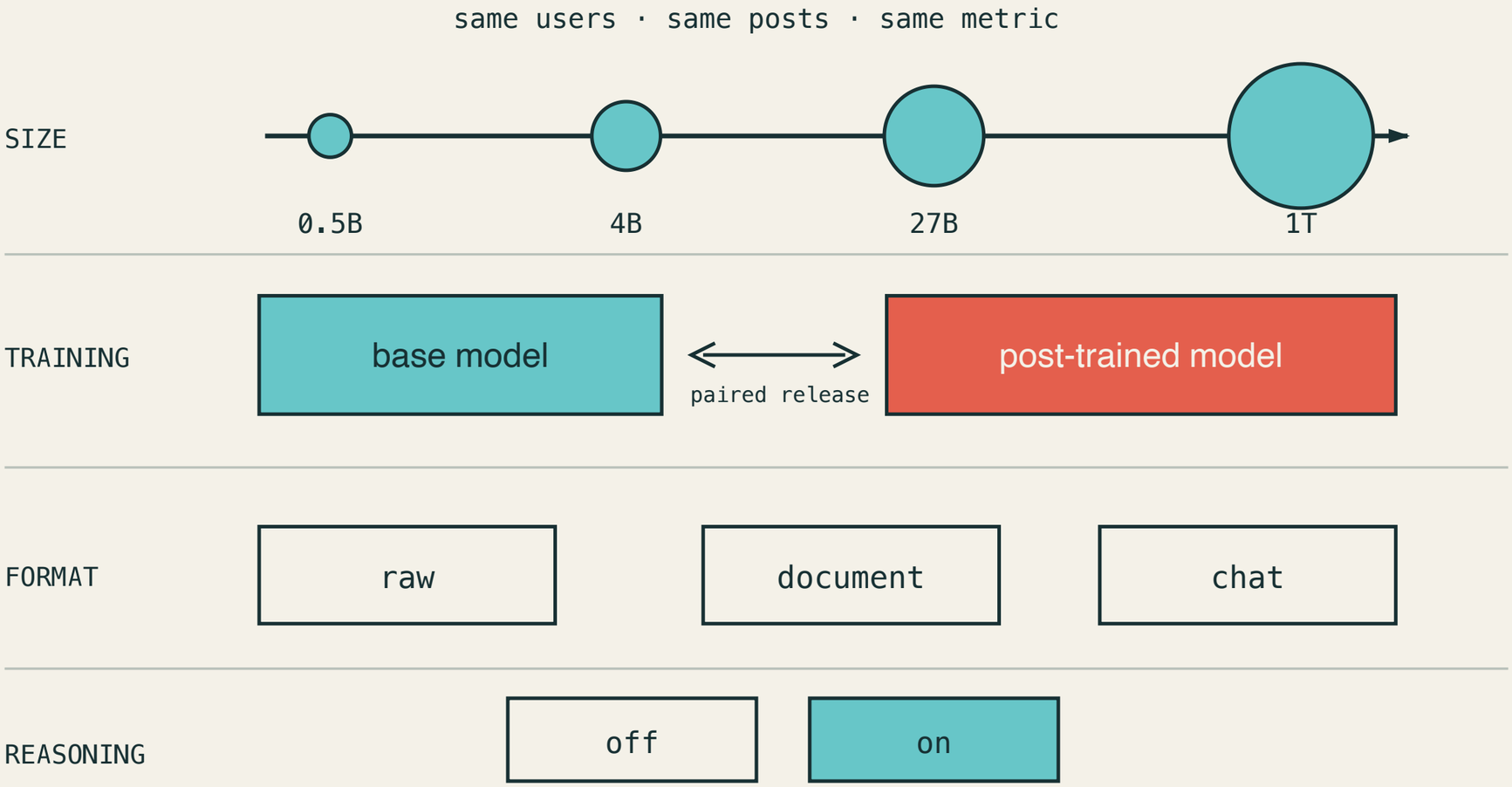

CAROLI LINNÆI REGNUM ANIMALE.
I QUADRUPEDIA
II AVES
III AMPHIBIA
IV PISCES
V INSECTA
VI VERMES
PARADOXA

What does the model know?



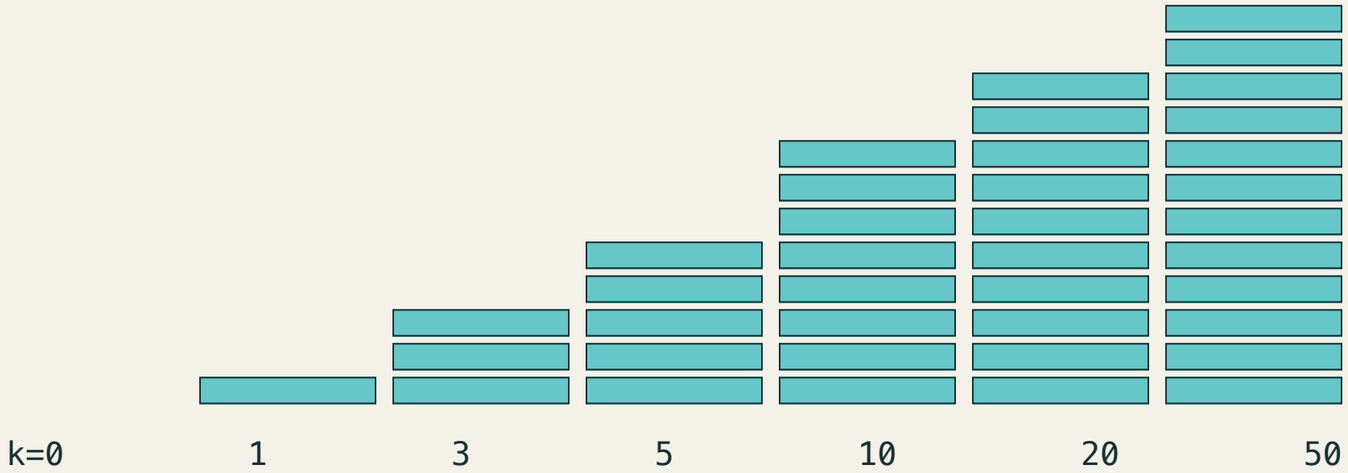
Description, demonstration, or environment?

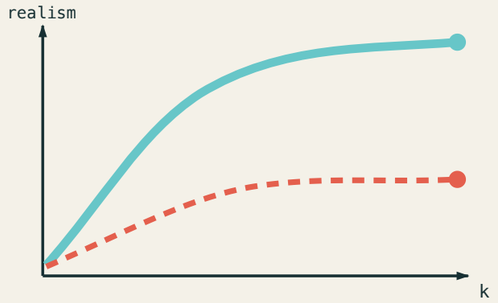
Models



Size × training × format × reasoning.

Dosed persona context





this user another user (control)

2 platforms × 2 eras

	BLUESKY	REDDIT
OLD SEP 2023–FEB 2024	**10,000** users **50,000** posts	**10,000** users **50,000** posts
NEW MAR–MAY 2026	**9,977** users **49,885** posts	**10,000** users **50,000** posts

Scale

34

base models

71

post-trained models

18

persona approaches

417M

post scores

8,619 experiments

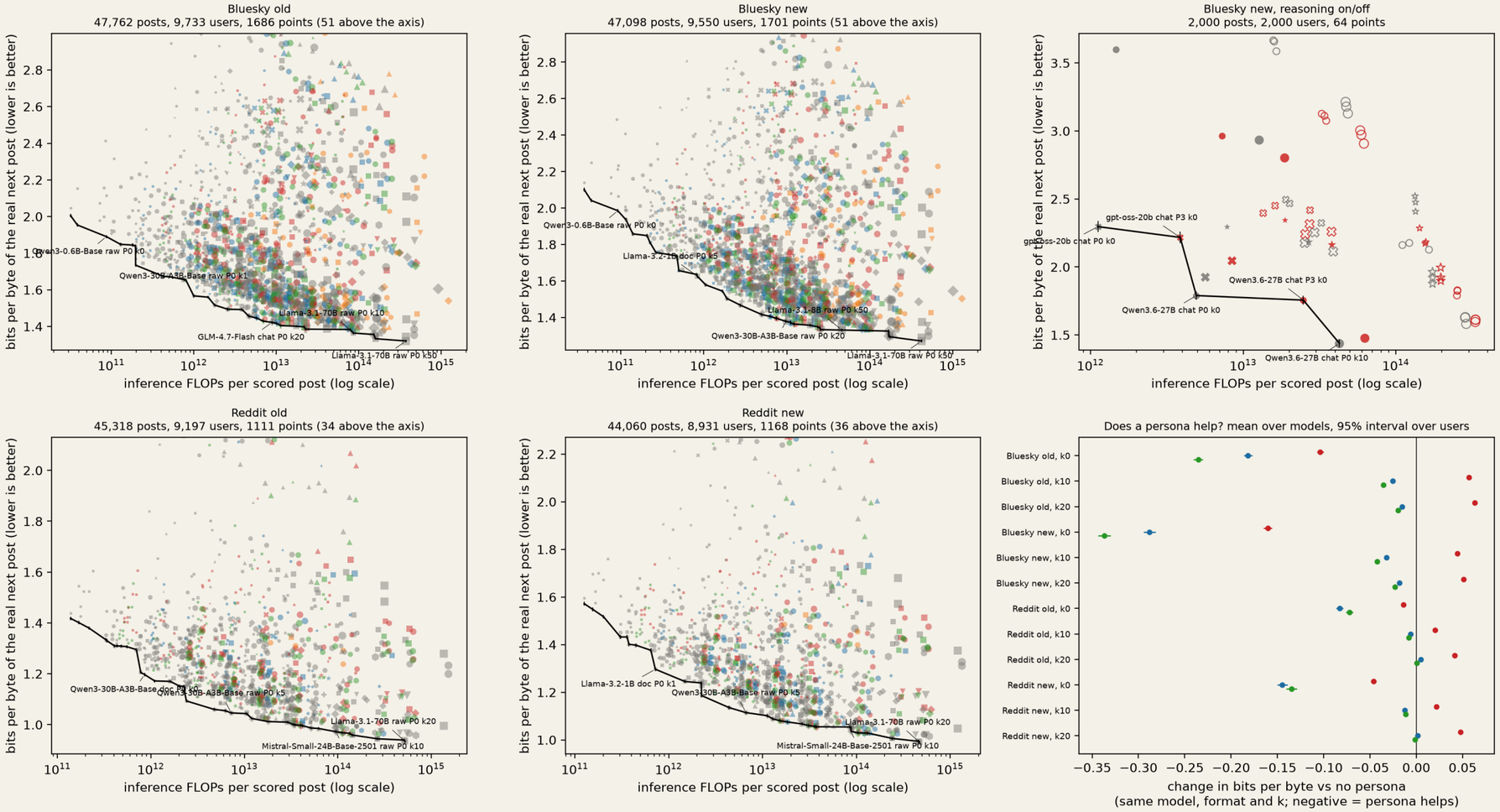
>8,746

GPU-hours

recorded lower bound

Compute × context

lower = better



FLOPs per post: forward pass counted per architecture from config.json and safetensors shapes (analysis/flops_model.py): 2 x active non-embedding weights per token (MoE: router + top-k routed + shared experts), logits at every position, causal attention per query-key pair with sliding/chunked/top-k windows, linear-attention layers linear in length; each post's own length; reasoning traces decoded token by token; no prefix caching. Every point in a panel is on the same posts. Black line: Pareto frontier (lowest loss at or below each FLOPs level), bars 95% intervals over users. Reasoning panel: traces capped at 2,048 tokens, 4 samples at T0.6/T1.0 averaged per post. Old = target posts 2023-09-01..2024-01-31, new = 2026-03-01..2026-05-11, on both platforms. Bluesky old users are early adopters (first 100k accounts) with at most ~7 months of earlier posts; Reddit old users have years of earlier posts.

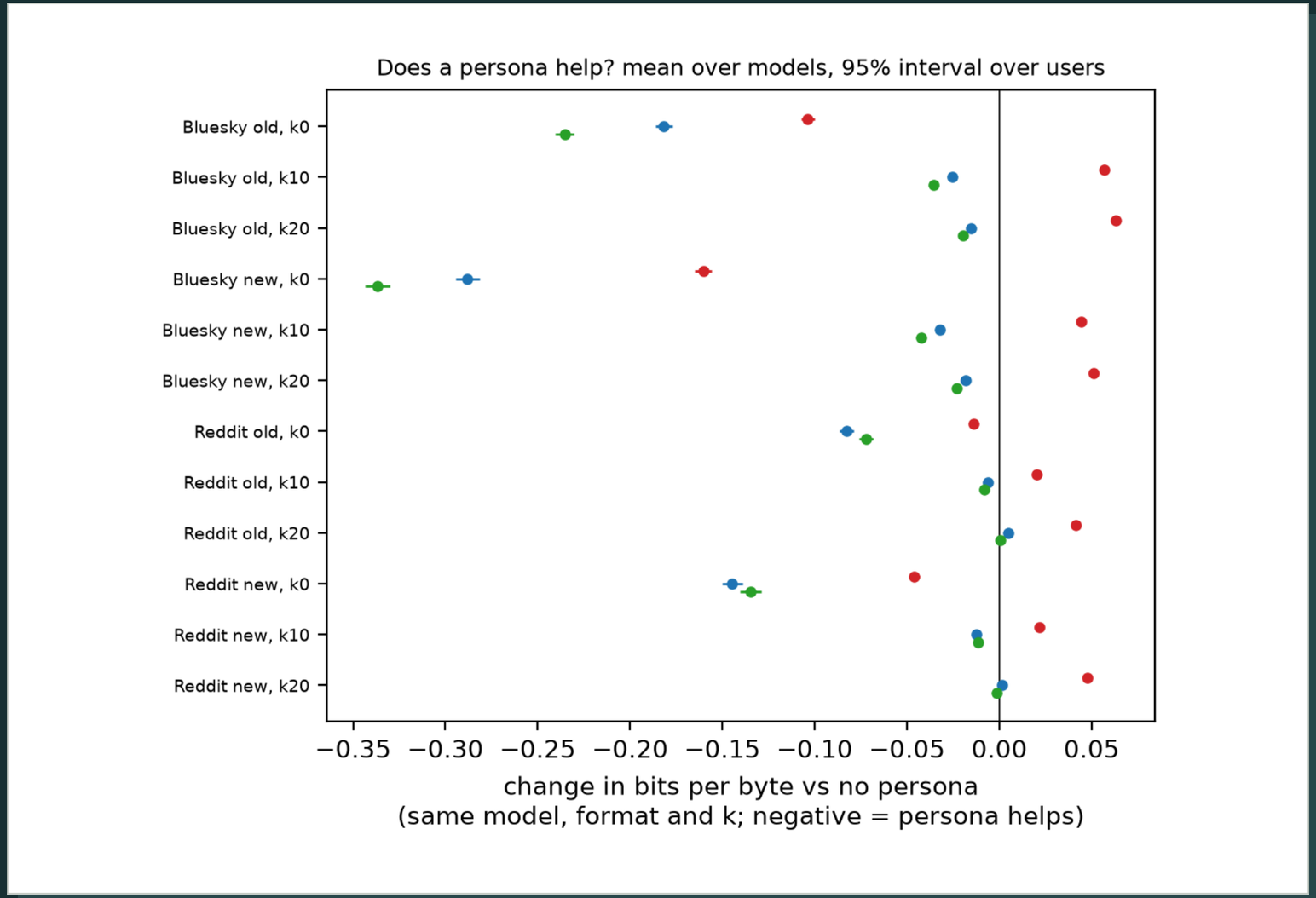
Show user *traces.*

P8 · older own posts

P7 · 3 representative posts

P3 · LLM biography

posts > descriptions



How do we leverage this for training?

